

Selective Semantic Correspondence under Real RGB-D Domain Shift: Separating Candidate Availability from Pose Recovery

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Abstract

Pose-free correspondence across independently captured RGB-D scans is often evaluated as if semantic matching and rigid registration were the same problem. We separate these objectives. Our method concatenates shallow, middle, and deep frozen DINOv2 patch tokens, retains top- K reference candidates, and uses 3-D spatial-consensus RANSAC to identify globally coherent correspondences without using the query-to-reference alignment during matching. On the original 14-pair matched subset, the ViT-G point estimate of retained-set semantic accuracy rises from 85.1% to 86.9% over ViT-L/14, but the pair-bootstrap interval includes zero; the expanded 16-source ViT-G audit reaches 85.0% while explicit candidate-set and wrong-pose failures remain. On one held-out target, ViT-G reaches 68.0% inlier coverage, 93.5% semantic inlier accuracy, and 1.54° rotation error. A source-only residual gate fit to a requested 90% retained semantic accuracy on source queries retains 22.1% of target queries at 96.6% retained semantic accuracy. Its target semantic mIoU is 89.5% conditional on retained queries, versus 19.7% over the fixed query population when abstentions count as false negatives. Cross-fitted pooling over all 16 held-out source folds retains only 9.1% of queries at 76.6% retained semantic accuracy, so the source-fit operating point is not a universal guarantee. We argue that correspondence quality, pose recovery, and abstention coverage should be reported separately, and make the failure taxonomy part of the result.

1 Introduction

Matching observations from independently captured RGB-D scans is useful for semantic map transfer, object retrieval, and 3-D scene understanding. The problem is deceptively easy to overstate: a correspondence can have the right semantic category while supporting a wrong global rigid transform, and a correct transform can have too few semantically useful matches. We therefore ask a narrower and testable question: *can a frozen visual descriptor plus spatial verification expose a useful subset of cross-scan semantic correspondences with high retained semantic accuracy?*

We study this question on annotated 3RScan reference/rescan pairs [2]. The method uses no known alignment while selecting candidates. It concatenates tokens from three transformer depths of a frozen DINOv2 back-

bone [1], keeps top- K reference candidates per query, and selects a spatially coherent hypothesis with 3-point RANSAC [3]. The resulting residual and inlier status provide an abstention mechanism. We evaluate semantic and instance label transfer separately from rigid pose error, and compare with classical FPFH [4] and learned FCGF [5] geometry-only baselines.

Figure 1 summarizes the matching path and makes the separation between matching-time inputs and evaluation-only alignment explicit.

Our central finding is deliberately asymmetric. ViT-G gives a higher point estimate of final retained-set semantic accuracy on the matched subset, but the interval includes zero and the experiment does not isolate a pre-consensus candidate-layer scale effect. The complete source audit contains pairs where even the most spatially consistent candidates are wrong. The strongest claim supported by the evidence is therefore *selective semantic correspondence*, not universal pose-free registration.

Positioning. Our setting sits between frozen visual representation learning and classical 3-D registration. DINOv2 provides the appearance descriptor, while FPFH and FCGF represent geometry-only local descriptors; the latter is a learned baseline trained for geometric matching rather than semantic identity [4, 5]. We focus on the evaluation gap that remains after candidate retrieval: semantic correctness, rigid-pose validity, and abstention coverage are measured as separate outcomes rather than collapsed into one registration score.

Our contributions are:

1. a pose-free top- K candidate and spatial-consensus protocol that separates semantic correspondence from global pose recovery;
2. a controlled ViT-L to ViT-G scale-up on an original 14-pair subset, followed by a 17-pair ViT-G audit, and classical FPFH plus learned FCGF geometry-only controls;
3. a source-only risk-coverage and empirical source-fit gate audit with explicit candidate-availability, wrong-consensus, and wrong-pose failure cases; and
4. a downstream cross-scan semantic/instance label-transfer evaluation with class-balanced mIoU.

2 Related Work

Learned 3-D correspondence and registration.

Geometry-first methods such as 3DMatch [6] and FCGF [5] learn local descriptors for partial-scan matching, while PointDSC [7] and GeoTransformer [8] use spatial structure to improve outlier rejection or registration. REGTR [9] further predicts correspondences with a transformer rather than relying on a hand-designed matching stage. These methods primarily optimize geometric inlier quality or registration recall. We instead use frozen RGB tokens to test semantic candidate quality under scan shift, and treat pose recovery as a separate outcome. The FPFH and FCGF rows in our experiments therefore serve as geometry-only controls, not as claims of state-of-the-art geometric registration.

Recent 3-D foundation models make the boundary worth stating explicitly: DUST3R [10] and MAST3R [11] infer pointmaps or image-level matches, while VGGT [12] predicts scene geometry and 3-D tracks from RGB views. Their inputs, outputs, and standard protocols differ from our native-coordinate RGB-D, post-hoc semantic-label-transfer setting, so we do not present them as directly comparable baselines. Semantic-aware registration methods such as SemReg [13] and ML-SemReg [14] are also adjacent: they train semantic/geometric registration models, whereas our descriptor is frozen and labels are used only for evaluation.

Visual features and semantic transfer. Large self-supervised visual features such as DINOv2 [1] provide strong appearance representations without task-specific correspondence training. Our use of DINOv2 is intentionally frozen: the contribution is the candidate-consensus evaluation protocol and its selective operating point, rather than a new descriptor-training objective. In particular, semantic agreement can remain high when a globally incorrect pose is selected, so a registration recall number alone is insufficient for the cross-scan label transfer setting studied here.

3 Method

3.1 Multi-scale candidate descriptors

For each RGB-D frame, we resize RGB with bilinear interpolation and depth with nearest-neighbor interpolation to 224×224 pixels, scale the complete intrinsic rows accordingly, and convert depth to meters. We sample the first 20 frames in manifest order from each side (the same 20-reference/20-rescan protocol for every headline pair). At each 14-pixel patch center with finite, positive depth, we back-project into the native scan coordinate frame and retain the point even when its annotation is missing. No 3-D point deduplication is applied across frames. RGB is ImageNet-normalized before frozen DINOv2 inference. For ViT-G/14, each selected block token is ℓ_2 -normalized, the

(2, 20, 40) tokens are concatenated, and the concatenation is normalized once more. Let q_i denote a query patch descriptor and r_j a reference descriptor. We retain the top- K candidates

$$\mathcal{C}_i = \text{TopK}_j \ q_i^\top r_j,$$

with $K = 20$. The alignment between the two scans is not used in this stage.

3.2 Spatial consensus

Let $x_i \in \mathbb{R}^3$ be a query point in native rescan coordinates and $y_j \in \mathbb{R}^3$ a reference point in native reference coordinates. The transform $T_{qr} = (R, t)$ maps query to reference coordinates. For a candidate set $\mathcal{C}_i$ of reference indices, we define

$$\hat{j}_i(T) = \arg \min_{j \in \mathcal{C}_i} \|Rx_i + t - y_j\|_2,$$

with the smallest reference index breaking ties. A sampled set of three distinct query indices is drawn without replacement; the three candidate slots are drawn independently and may refer to the same reference point. We estimate the hypothesis with the SVD Procrustes fit and correct a reflected solution by flipping the last singular vector so that $\det R = +1$. The implementation has no separate non-collinearity rejection; degenerate samples therefore follow the same SVD fit and are scored normally. The hypothesis is scored by assigning each query to its nearest candidate in $\mathcal{C}_i$ after transformation:

$$s(T) = \sum_i \mathbf{1} \left[\left\| Tx_i - y_{\hat{j}_i(T)} \right\|_2 \leq \tau \right], \quad \tau = 0.10 \text{ m.}$$

The first maximum-scoring hypothesis wins ties by iteration order. The final transform is refit once on its inliers, then candidates are reassigned once; there is no further alternating optimization. Nearest assignment permits multiple queries to select the same reference point. A query is accepted by the basic protocol when its residual is at most τ ; more conservative selectors retain a subset by residual or use a source-fit residual gate.

3.3 Metrics and split

Semantic and instance correctness compare the selected reference labels with the query annotations only after matching. We call this retained-set semantic accuracy (equivalently, micro precision on the retained set), and use that name throughout. Matching uses every depth-valid sampled point; non-zero query instance and semantic IDs define the post-hoc evaluation population, and a selected reference point with a missing label is counted as incorrect. Coverage is the fraction of that fixed query population retained. Pose error is scored against the hidden 3RScan transform only after matching. The frozen headline subset contains 17 availability-selected reference/rescan relations with complete RGB-D sequences and annotation meshes: 16 source pairs and one target pair, totaling 15,048 source

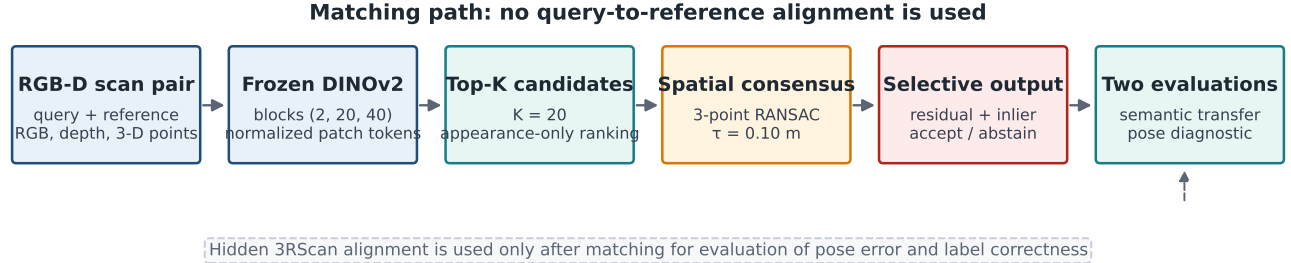


Figure 1: The pose-free matching path. Frozen multi-scale tokens produce a top- K candidate set, spatial consensus selects a coherent hypothesis, and residuals expose a selective accept/abstain decision. The hidden 3RScan alignment is used only after matching for evaluation; annotation availability is not used to construct candidate sets.

and 1,069 target query records. The original target relation, 02b33dfb/fcf66d9e, remained held out when three source relations were added. This is not a random sample of the 1,004 relations listed in 3RScan metadata; the exact scan IDs and inclusion scope are given in the supplement. All 34 scan UUIDs are distinct across the selected relations, but distinct IDs do not establish that their physical capture sites are independent. The target is a single held-out diagnostic, not evidence of broad generalization; all thresholds and selective choices are fit source-only.

For a selected set A , the reported conditional class-balanced mIoU averages IoU over the union of non-zero query and selected-reference labels present in A , with zero-union classes omitted. It is therefore a conditional utility metric, not a full-query segmentation score. We additionally report in the supplement a fixed-query, abstention-as-false-negative mIoU: the class set is fixed from all non-zero query labels in the split, rejected queries predict a void label, and zero-union classes are assigned IoU zero. Instance labels are the 3RScan annotated mesh’s shared ‘globalId’ values, not side-specific ‘objectId’ values.

With $T_{gt} = (R_{gt}, t_{gt})$ in the same query-to-reference direction, pose errors are

$$\theta = \arccos\left(\text{clip}_{[-1,1]} \frac{\text{tr}(RR_{gt}^T) - 1}{2}\right), \quad e_t = \|t - t_{gt}\|_2,$$

where $\theta \in [0, \pi]$ is reported in degrees. Strict pose success means $\theta < 5^\circ$ and $e_t < 0.25$ m; both inequalities are required.

For downstream label transfer, the selected reference semantic or instance label is copied to the query patch. We report micro accuracy and class-balanced mean IoU over the retained patches. This evaluates utility of the correspondence output without claiming a full VLM or runtime improvement.

Table 1: Main results. Retained semantic accuracy is measured on accepted inliers; FPFH is a non-equal-density diagnostic and FCGF is a geometry-only control. A dash denotes that no single aggregate rotation/translation error is meaningful for a multi-pair row.

Setting	Coverage	Sem. acc.	Rot. err.	Trans. err.	Strict pose
ViT-L target	65.1%	92.4%	3.97°	0.13	1/1
ViT-G target, $K = 1$	27.9%	92.3%	–	–	1/1
ViT-G target	68.0%	93.5%	1.54°	0.17	1/1
ViT-G, 16 source	30.4%	85.0%	–	–	1/16
FPFH one-way, all 17	4.31%	67.5%	–	–	0/17
FCGF top-20, all 17	18.7%	53.3%	–	–	0/17
Permuted DINO reference, 16 source	10.5%	46.3%	–	–	0/16

4 Experiments

4.1 Backbone scale-up and geometry control

The main protocol uses ViT-G/14 with blocks (2, 20, 40), $K = 20$, $\tau = 0.10$ m, and 1,500 RANSAC iterations. The ViT-L comparison uses its structurally matched (2, 12, 24) blocks. The classical geometry-only control uses 0.08 m voxelized points, FPFH with 0.16 m normal and 0.40 m feature radii, and 5,000 RANSAC iterations. The learned FCGF control uses the published 3DMatch checkpoint, 0.025 m voxelization, the same 14-pixel point sampling and top- K consensus protocol, and 3,000 RANSAC iterations. Both controls use the same annotated pair split.

On the original matched 14-pair subset, ViT-G gives a higher weighted retained semantic accuracy point estimate, 86.9% versus 85.1% for ViT-L; a pair-bootstrap interval for the +1.8-point difference is $[-0.4, +5.3]$ points, so we treat this as descriptive scale evidence rather than a superiority claim. The expanded 16-source ViT-G audit reaches 30.4% weighted coverage and 85.0% weighted retained semantic accuracy. On the target, coverage improves from 65.1% to 68.0%, retained semantic accuracy from 92.4% to 93.5%, and rotation error from 3.97° to 1.54°. The top- K mechanism is consequential rather than cosmetic: relative to the same ViT-G protocol with $K = 1$, $K = 20$ raises target coverage from 27.9% to 68.0% while

retained semantic accuracy rises from 92.3% to 93.5%; on the source pool it trades 11.2%/89.5% for 30.4%/85.0% coverage/retained accuracy. The full FPFH diagnostic reaches 4.31% weighted inlier coverage and 67.5% retained semantic accuracy, with no strict pose success under the $5^\circ/0.25$ m criterion; its voxelized query denominator is not comparable to the patch denominator. The learned FCGF control reaches 18.7% coverage and 53.3% retained semantic accuracy with 0/17 strict pose successes. As a negative control, randomly permuting reference descriptor assignments while keeping the same points and consensus budget reduces the source result to 10.5% coverage and 46.3% retained semantic accuracy, with 0/16 strict pose successes; the held-out target falls to 7.9%/3.6% and 162.2°.

Relation-level uncertainty. We bootstrap the 16 source relation pairs rather than individual patches. The weighted retained semantic accuracy is 85.0% with a 95% pair-bootstrap interval of [69.5%,93.5%], weighted coverage is 30.4% [20.1%,41.0%], and strict pose success is 6.3% [0.0%,18.8%]. The width of these intervals is itself evidence of heterogeneous relation difficulty; the single held-out target is reported as a point estimate. Because physical-site overlap between distinct scan UUIDs was not verified, these intervals quantify variation across the recorded relations and do not establish independent-site uncertainty. A post-hoc RANSAC seed audit keeps the target stable at 93.1–93.5% retained semantic accuracy and 67.6–68.2% coverage, but the source semantic aggregate ranges from 83.1% to 88.0% across three fixed seeds. The variation is concentrated in hard wrong-pose pairs, so the method should be interpreted as a selective verifier rather than a deterministic global-pose estimator.

4.2 Selective risk and failure taxonomy

Figure 2 summarizes the complete source and target query audit. At 20% retained coverage, low-residual selection has 87.4% source and 97.7% target retained semantic accuracy. A source-only residual gate of 0.040931 m, fit to a requested 90% retained semantic accuracy on source pairs, retains 22.1% of target queries at 96.6% retained semantic accuracy. A source-selected frame-level reciprocal/margin control reaches only 87.6% retained semantic accuracy at 12.1% target coverage, so descriptor confidence alone does not explain the consensus result. No selector reaches a 95% retained-semantic-accuracy target on the full heterogeneous source pool. Leave-one-pair-out source-fit gates at the same requested 90% retained semantic accuracy on source pairs still yield 0.0% and 10.0% retained semantic accuracy on held-out 4aca and 95be, respectively. A post-hoc candidate-set audit shows that a correct semantic label appears in the top-20 set for only 19.5% of 4aca queries but 69.2% of 95be queries; the corresponding correct-inlier recoveries are 0.0% and 0.7%. Thus a residual gate cannot repair 4aca’s candidate omission, while 95be is dominated

by wrong-consensus/pose selection even when a correct semantic candidate is often available. Cross-fitted pooling over all 16 held-out folds retains 1,364/15,048 source queries (9.1%) at only 76.6% retained semantic accuracy; pair heterogeneity means the source-fit 90% target is not a universal guarantee. The 0.03304 m source-fit selector retains 14.1% of target queries at 98.7% retained semantic accuracy, but its raw 90.1% Clopper–Pearson lower bound is not a family-wise guarantee after searching many residual prefixes. A fixed 0.02 m audit gate has an unadjusted 90.9% source lower bound at only 1.9% source coverage, and transfers to 3.3% target coverage at 100% retained semantic accuracy; the formal Bonferroni audit finds no 90% family-wise point.

The failure taxonomy is important. In 4aca, only 19.5% of evaluated queries have any semantically correct top-20 candidate, and no correct semantic inlier survives; it is a direct candidate-set failure. In 95be, 69.2% have a semantically correct candidate, but only 1.0% of those candidate-hit queries become correct inliers, exposing a wrong-consensus/pose failure. Other pairs, including 20c993 and 6bde, can have high-semantic inliers while selecting an incorrect global pose. These cases motivate reporting candidate availability, semantic correctness, pose error, and abstention coverage as separate axes. Figure 3 visualizes these failure families over the expanded source audit.

4.3 Cross-scan label transfer

Table 2 treats each selected match as a semantic and instance label-transfer prediction. On the target, consensus inliers provide 85.5% semantic mIoU at 68.0% coverage. The source-fit residual selector raises conditional semantic mIoU to 89.5% and conditional instance mIoU to 90.7% at 22.1% coverage. Over the fixed target query population, with abstentions counted as false negatives, its semantic mIoU is 19.7%. The corresponding pooled conditional semantic mIoU is 61.2%, while the pair-macro conditional semantic mIoU is 67.3%; the two values differ because the former pools retained queries before taking the class mean and the latter first computes one mIoU per pair. Both are conditional on retention and are reported to avoid conflating them. The downstream geometry-control audit in the supplement gives FCGF 64.2% semantic mIoU at 17.5% target coverage and FPFH 82.8% at 13.2%; FPFH uses voxelized points, so this is a diagnostic comparison rather than an equal-density claim.

5 Limitations and conclusion

This is a selective correspondence study, not a universal registration or efficiency claim. The frozen headline snapshot contains 17 availability-selected reference/rescan relations (16 sources and one target), out of 1,004 relations listed in the metadata. The exact IDs and inclusion scope are reported in the supplement. Although scan UUIDs are dis-

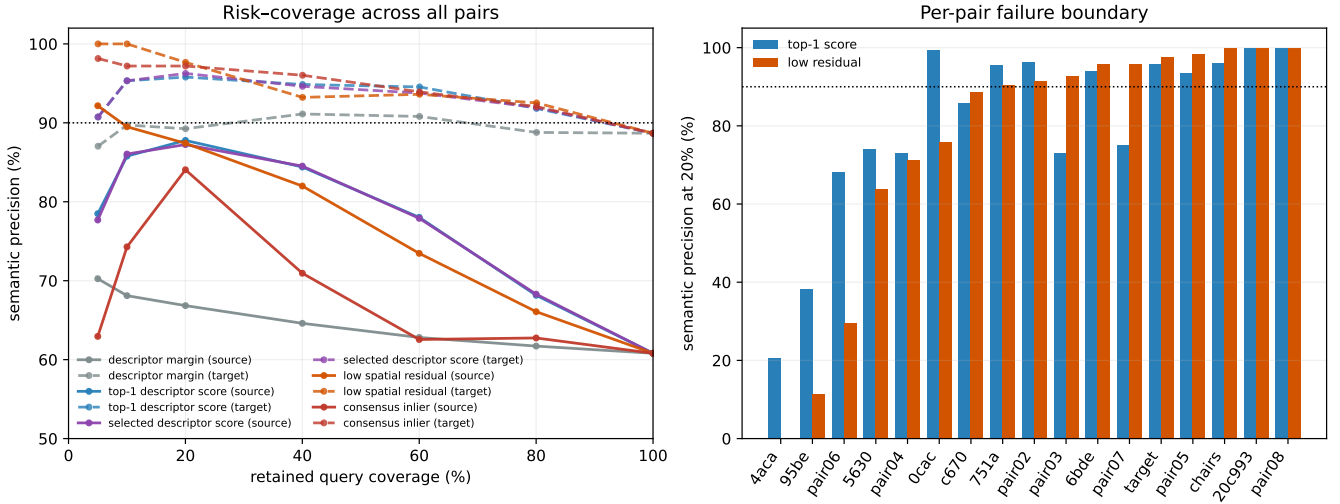


Figure 2: Full selective audit. Left: source (solid) and target (dashed) risk-coverage curves. Right: per-pair retained semantic accuracy at 20% coverage; the low bars for 4aca and 95be are candidate-availability/consensus failures, not merely aggregate pose failures.

Table 2: Cross-scan label transfer. Conditional mIoU is class-balanced over retained queries; the fixed-query abstention-aware version is in the supplement.

Dataset / selector	Cov.	Sem. mIoU	Inst. mIoU
Source / all	100.0%	19.4%	19.3%
Source / consensus	30.4%	41.5%	39.0%
Source / source-fit	8.9%	61.2%	51.5%
Target / all	100.0%	67.4%	55.5%
Target / consensus	68.0%	85.5%	72.3%
Target / source-fit	22.1%	89.5%	90.7%

tinct, physical-site overlap across relations was not verified; pair-bootstrap intervals therefore describe relation-level variation rather than confirmed independent-site uncertainty. One held-out target cannot establish broad generalization. Applied without refitting, the source-only residual gate is not a guarantee outside this split; the present snapshot does not claim external validation beyond the reported target. FPFH and FCGF are geometry controls; neither receives RGB evidence, and the FCGF result is reported as a control rather than tuned for 3RScan. The current label-transfer task demonstrates utility; downstream VLM task success and sparse-execution speedups remain future work.

The evidence supports a narrower conclusion: the larger frozen transformer has a higher final retained semantic-accuracy point estimate on the matched subset, but the uncertainty interval includes zero and no pre-consensus candidate-layer effect was isolated. Spatial consensus makes a useful high-accuracy retained subset explicit, but neither component eliminates candidate-availability, wrong-consensus, or wrong-pose hypotheses. Evaluating these axes separately yields a more reliable and reproducible account of cross-scan correspondence under real RGB-D

shift.

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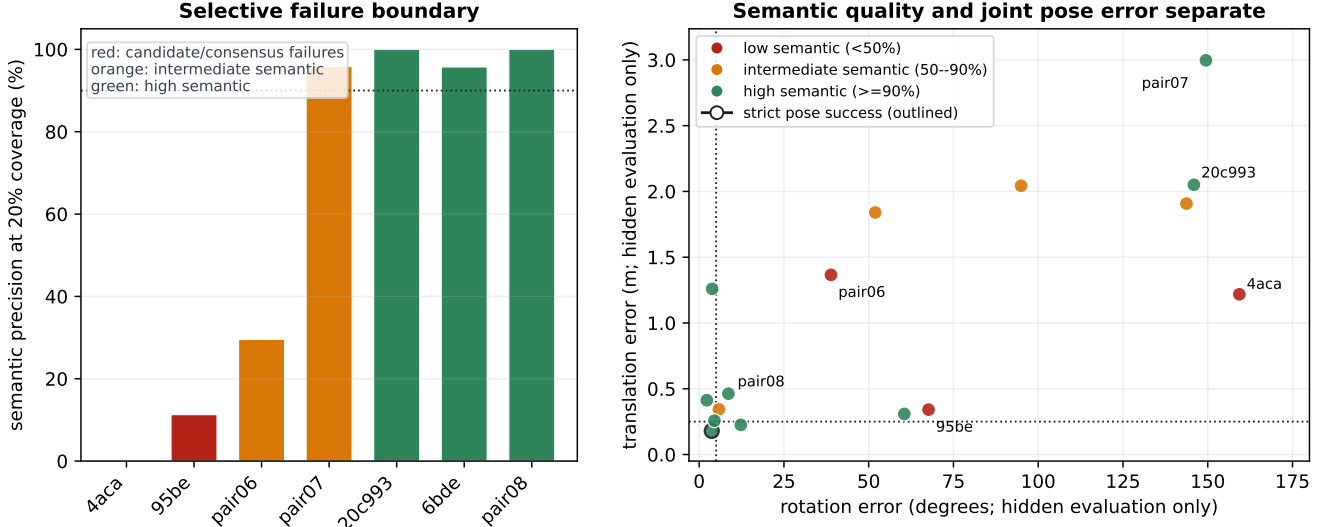


Figure 3: Joint failure audit on source pairs. Left: at fixed 20% coverage, candidate-availability/consensus failures remain low even after residual selection. Right: rotation and translation errors are shown jointly; dashed lines mark the strict $5^\circ/0.25$ m pose gate. Colors are mutually exclusive retained-semantic-accuracy bands, and marker outlines identify strict pose successes, so a high-semantic but wrong-pose pair such as 6bde is visible.

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